

Spectra Sleuth Showdown

Two white lights, two hidden color fingerprints

At a glance: Can two white lights hide different color fingerprints? Plan about 80 minutes for the core block; 4 to 6 teams of 4 to 5 students.

LEARNING OBJECTIVES

By the end of this activity, each student can:

- Explain the core science of this challenge: diffraction, spectra, light fingerprints, fireworks colors without flame (Understand).
- Use reference spectra to identify and match mystery light-source patterns (Apply).
- Analyze whether a spectrum is a smooth rainbow, broad band, or set of separate lines (Analyze).
- Defend each source match with a labeled sketch and visible spectral evidence (Evaluate).

MATERIALS FOR 25 STUDENTS AND 6 TEAMS

- diffraction glasses or gratings
- white LED and neon or gas sources
- incandescent source if available
- spectrum cards
- museum clue sheets

PREP BEFORE STUDENTS ARRIVE (15 TO 20 MIN)

- 01 Stage verified light sources in a dimmable viewing area and place reference cards, clue sheets, and sketch sheets at team stations.
- 02 Stage the scoring sheet and confirm that at least one true neon or other verified line source is available.
- 03 Allow no flames, lasers, or direct sun viewing. Diffraction glasses are not eye protection; only staff handle plugged-in or hot light sources.
- 04 Load the activity slides and ready the projected timer.

MINUTE-BY-MINUTE FACILITATION

Phase 1 Hook

0:00 - 0:05

Pose the mission as a challenge: "Can two white lights hide different color fingerprints?" Take a quick prediction by show of hands.

Phase 2 Prediction

0:05 - 0:12

Before viewing each mystery source, teams predict whether they expect a smooth rainbow, broad band, or separate lines.

Phase 3 Constraints

0:12 - 0:18

Observe from the marked safe position, sketch before matching, and cite a visible pattern. No flames, lasers, or direct sun viewing.

Phase 4 Observe and match spectra

0:18 - 0:44

Teams observe, sketch, match, and justify each mystery spectrum. Circulate without naming a source before its sketch is complete.

Phase 5 Test and record

Teams compare each sketch with the reference cards and record a source match plus the visible evidence.

Phase 6 Evidence check

Teams audit their matches, revise only when the observed pattern supports the change, and prepare a fireworks-science explanation.

Phase 7 Score, debrief, cleanup

last 12 min

Teams submit a score slip, answer a debrief question with evidence, return cards and gratings, and leave light sources for staff to unplug.

TROUBLESHOOTING

Problem: A spectrum looks washed out. **Fix:** Dim ambient light, increase viewing distance, and use a clear view of the source.

Problem: Every source looks like a broad band. **Fix:** Confirm that the line source is true neon or another verified gas source, not an LED night light.

Problem: Sketches disagree. **Fix:** Reobserve from the same marked position and compare pattern type and line locations, not brightness alone.

Problem: A team finishes early. **Fix:** Give an additional clue card or ask them to connect one line pattern to an emitting element.

DIFFERENTIATION: THREE-TIER SUPPORT PROMPTS

Tier 1 (stuck at the start):

- "What is the one science idea behind this challenge?" Point them to diffraction splits light.

Tier 2 (matching but not reasoning):

- "Which visible pattern supports your source match?"

Tier 3 (ready to extend):

- "Compare a smooth, broad-band, and line spectrum and explain what each reveals about its source."

DEBRIEF QUESTIONS (PICK 2 TO 3)

- How can two white lights have different spectra?
- What does a line spectrum tell you?
- How do chemists make firework colors?

SCORING RUBRIC (100 POINTS)

SCORING CRITERION	POINTS
Correct matches	40
Spectrum sketches	20
Explanation	20
Exhibit connection	20
Total	100

CLEANUP AND SOURCES

Return gratings, reference cards, and clue sheets; staff unplug and cool light sources; then collect score slips.

SOURCES Science History Institute Flash Bang Boom; Exploratorium Spectra